

- ```
int x;
x = ();
```

## Enzymen

✓ middle = Monday;

3                      4  
↑                      ↑  
Tuesday, Wednesday

$\rightarrow$  Monday, Tuesday = 5, Wednesday



```
typedef int yash;
```

```
yash x, y, z; ✓
int c0; ✓
```

```
typedef void no_data;
```

```
typedef int marks;
```

```
marks sub1, sub2, sub3;
 ↑ ↑ ↑
```

```
sub1 = 72;
```

```
sub2 = 55;
```

⚡

```
⇒ struct stu_data
{
 int rno;
 float result;
};
```

```
struct stu_data student1;
 ↑ ↑
data type var-name
```

```
typedef struct stu_data
```

```
{
 int rno;
```

```
⇒ float result;
```

```
} s_data;
```

```
s_data student1;
 ↑ ↑
data type var-name
```

## C Tokens

- ① keywords  $\rightarrow$  32 keywords (ANSI) (break, void, switch, int)
- ② Identifiers :- var-name, function-name, name\_array
- ③ Constants
- ④ Strings
- ⑤ Operators
- ⑥ Special Symbols :- {}, [], ...

int x;2

x;2

P1  
P2

- int  
1 x  
& x

x = 7.9;  
y = 037;  
z = 0x2F;

## Constants

- ① Numeric Constants

$\rightarrow$  int  
 $\rightarrow$  real

w = 32

w = 32L  
a = 55.0L

- ② char constants

$\rightarrow$  single char  
 $\rightarrow$  string

char = "abc";

a | b | c | \0

'y' vs "y"  $\rightarrow$  2 bytes  
y \0

-4.5e-4

1.5E+7

1 byte

①

Back slash Char Constant

\n

\t

\0  
"

printf ("\new line");

x = printf ("new line");

"ABC"

printf ("ABC"); → printf (" | "ABC | " ");

y = scanf ("%d %d", &a, &b);  
↑ ↑

## \* Symbolic Constants:-

#define      NAME      VALUE

#define      MAX      100

#define      PI      3.14

{

int main()

{  
float area;

int count = 0;

while (count < <sup>100</sup>MAX)

{

area = <sup>3.14</sup>PI \* count \* count;

count++;

printf("x.p", area);

return 0;

}

## \* Const variable

int main()

{

const float PI = 3.14;

PI += 1; ✗

}

\* Overflow & Underflow:-

→ Assume int is of 1 byte ASSUME

|                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 7              | 6              | 5              | 4              | 3              | 2              | 1              | 0              |
| 2 <sup>7</sup> | 2 <sup>6</sup> | 2 <sup>5</sup> | 2 <sup>4</sup> | 2 <sup>3</sup> | 2 <sup>2</sup> | 2 <sup>1</sup> | 2 <sup>0</sup> |
| 128            | 64             | 32             | 16             | 8              | 4              | 2              | 1              |

-128 to 127

IP

127 → 0 1 1 1 1 1 1 1

128 → 1 0 0 0 0 0 0 0

129 → 1 0 0 0 0 0 1

128

127

-128

-127

!